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KHOA CÔNG NGHỆ THÔNG TIN

VNUHCM - UNIVERSITY OF SCIENCE

FACULTY OF INFORMATION TECHNOLOGY

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PROJECT PROPOSAL

& PLAN

Instructor: Nguyễn Lê Hoàng Dũng

Nguyễn Minh Huy

Class: 24C05

Team: 08

24C05

Ho Chi Minh City

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1 Member Contribution Assessment

[List every member. Contribution (%) must add up to 100%. Evidence: links to tasks on the board, commits or pull requests.]

ID	Name	Role(s)	Contribution (%)	Evidence
24127278	Hoàng Minh Huy	<Team lead / PM>	22	https://github.com/Ne1ntendo/InfraLine/issues/6
24127048	Nguyễn Thế Huy	<Business analyst>	22	https://github.com/Ne1ntendo/InfraLine/issues/8
24127450	Nguyễn Chí Minh	<Designer>	17	https://github.com/Ne1ntendo/InfraLine/issues/9
24127329	Lê Nguyễn Quốc Bảo	<Developer>	22	https://github.com/Ne1ntendo/InfraLine/issues/10
24127142	Đoàn Minh Tuấn	<Tester>	17	https://github.com/Ne1ntendo/InfraLine/issues/7

2 Vision

2.1 Preliminary Problem Statement

When workplace infrastructure breaks down, such as broken machinery in a factory or a faulty projector in a school, users need to report these issues to upper management, who then need to assign maintenance personnel to resolve those issues. To streamline this process, a centralized **work order management system** is needed. Such a system needs to allow users to report issues regarding broken equipment. These reports are then forwarded to the maintenance staff and management staff as work orders. The maintenance staff can view and carry out these work orders, while the management staff oversees the entire process and makes adjustments if necessary.

The software will be a web application built with Python, HTML, CSS, and JavaScript, with the data stored in PostgreSQL. When a worker or student comes across broken equipment, they can write a report on a forum. The report should include a description of the issue, where it happened, and photos of the equipment. The report will then be subject to approval by administrators. Once approved, it will be shown on the forum for everyone to see. It will also appear in a checklist visible only to maintenance and management staff, where it will be referred to as a **work order**. Apart from the title and description, the work order will have additional attributes such as status, location, priority, assignee(s), and type of equipment (mechanical, electrical, etc.). Administrators have full control over these attributes and can change them if needed. To get a better understanding of the problem, a technician may enter a private chat to discuss the issue with the report author. The maintenance staff can then carry out these work orders and update their status. When the work is finished and the issue is resolved, an automated email is sent to the report's author to inform them about it.

The documentation for the software should be adaptable to change. It should have a structural order with bold headlines, and references to external sources should be cited accordingly.

2.2 Product Position Statement

Element	Content
For	Businesses and schools
Who	Need to manage workplace equipment maintenance
The (product name)	Work order management software
That	Streamlines the process of reporting and fixing workplace infrastructure
Unlike	Emails and messages
Our product	Focuses issue reports and maintenance work orders into a centralized system

2.3 Stakeholders and Users

#	Stakeholder / user	Description	Main interests
1	Workers/Students	Main beneficiaries of the equipment in the workplace.	• Reporting issues • Tracking issues
2	Technicians	Personnel whose job is to repair broken equipment.	• Viewing work orders • Carrying out work orders
3	Administrators	Upper management of the workplace.	• Managing work orders (see status, assign priority, set tags...) • Manage the system (accounts, locations...)

2.4 Scope and Key Features

ID	Feature	Description	Priority (M/S/C/W)
F01	Broken infrastructure report	Users can upload a post to report broken infrastructure on a forum for the maintenance team to see.	Must
F02	Communications channel	A maintenance team member can enter a private chatbox with the post author to get a better understanding of the problem through back-and-forth chatting.	Could
F03	Location ping location	Users can narrow down the location via a dropdown selection box where the broken infrastructure is located with a picture or short description of the fault.	Must
F04	Maintenance history	Users and administrators can see fixed infrastructure, who fixed it, and when the problem was dealt with.	Should
F05	Maintenance status	Users and administrators can see the current status of the problem, whether it's addressed, still being worked on, or no response.	Must
F06	Maintenance team checklist	Maintenance team members will have a checklist of broken infrastructure they are tasked to deal with.	Must

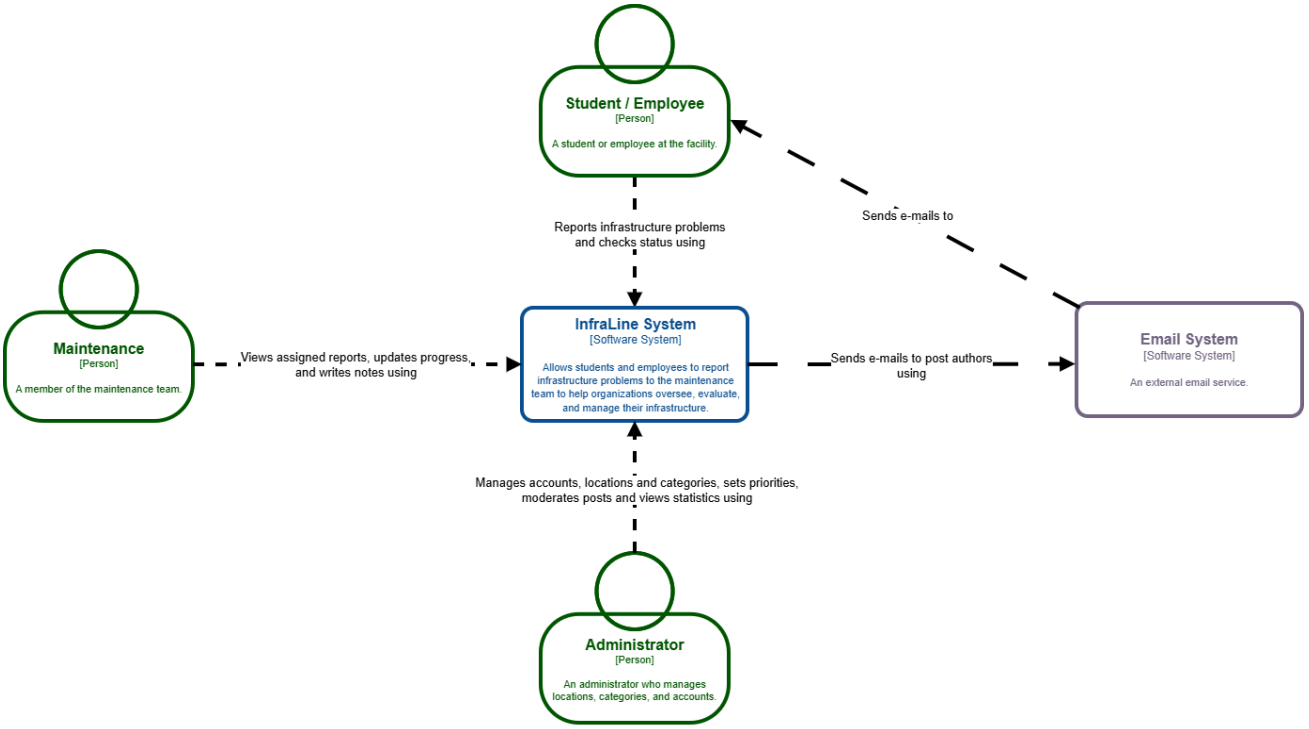
F07	Email response	An automatic email response will be sent to the post's author whenever the problem has been dealt with.	Could
F08	Broken infrastructure post tags	Users and administrators can apply tags to their posts to specify whether the broken infrastructure is mechanical, electrical, critical,... so the maintenance team has an easier time judging and prioritizing tasks.	Must
F09	Set task priority	Administrators can set priority to problems posted on the forum.	Must
F10	Problem moderation	Administrators can approve or reject a post if they feel the problem isn't worth addressing or is outside the scope of the maintenance team.	Must

Out of scope:

- ✔ Mobile support
- ✔ AI assistant
- ✔ 2D/3D floor map of the facility

3 Proposed Solution

3.1 Preliminary Architecture



System Context diagram for InfraLine

3.2 Technology Stack

Layer	Technology	Reason for choice
Front end	REACT	Our team has experience with REACT, through its components the front end will be easier to maintain and clean to look at.
Back end	Django	Easy URLs setup, page rendering through templates, built-in authentication with setup,
Database	PostgreSQL (Supabase)	Supabase provides 2 active projects on the free plan.
Hosting / deployment	Vercel, Render + Gunicorn	Vercel to host the front end, Render to host the back end with Gunicorn to load Django codes.
Testing & CI	GitHub Actions	The least setup with unmetered minutes for public repositories.

3.3 Hardware and Operating Environment

Development:

- Hardware: Windows laptop, computers.

- Languages: Python, Node.js, HTML, CSS, PostgreSQL.
- Code editor: VS Code / Pycharm.
- Repositories: Git and GitHub.
- Task board: GitHub Projects.

Testing:

- Local testing: Django's built in test runner.
- Unit testing: Python unittest library.
- Automated testing: GitHub Actions.
- Testing data: Seed data for testing, made by us to test our features.
- Devices and browsers: Chrome, Microsoft Edge, Firefox for desktops and laptops, Safari and Chrome for mobile devices.

Running:

- Front end: Vercel. With its Vercel's free "Hobby" plan, 100GB of fast data transfer and 10GB used on deployment amongst other free limits makes it suitable for this project.
- Back end: Gunicorn to load Django's codes, Render to host the back end, Gunicorn only runs locally on Linux, but through Render we can deploy it.
- Database: Supabase PostgreSQL, 2 free active projects, 1 GB file storage, and unlimited API requests on the free plan.
- Email sending: Resend or Mailjet

4 Project Plan

4.1 Process and Phase Gates

The project follows the waterfall model. Each phase ends with a phase gate (a PA submission). After a PA is submitted it becomes a baseline; later changes must be updated in the document and listed in the next PA.

PA	Phase	Due	Exit criteria (fill in)
PA1	Initiation & planning	End of week 3	Unify the idea, scope, tech stacks, risks, roles and planning of InfraLine.
PA2	Requirements	End of week 5	Ensure the project's use cases, output and implementation legality and follow the initial planning.
PA3	Design	End of week 7	Complete the code architecture, database structure and interface design.
PA4	Test planning	End of week 8	A complete test plan with written test cases, test methods and test environment.
PA5	Implementation, testing & deployment	End of week 10	Implement must-have features, test those features and release a demo.
Final	Presentation & demo	Week 11	Prepare slides, workflow diagrams and a presentation script.

4.2 Work Breakdown Structure (WBS)

1. **Initial planning**
   - **1.1. Idea specification**
     - 1.1.1. Idea proposing
     - 1.1.2. Problem specification
     - 1.1.3. Target users
     - 1.1.4. Core features
     - 1.1.5. Tech stacks
     - 1.1.6. Similar products finding
     - 1.1.7. Feasibility checking
   - **1.2. Team contracts**
     - 1.2.1. Main roles distribution

1.2.2. Meetings scheduling

1.3. Requirements

1.3.1. Define use cases

1.3.2. Ensure legality

1.3.3. Cost defining

2. Design

2.1. Back-end

2.1.1. Design code architecture

2.1.2. Design database structure

2.1.3. Design API routes (If necessary)

2.2. Front-end

2.2.1. Define necessary pages.

2.2.2. Design pages' UI

3. Implementation

3.1. Back-end

3.1.1. Implement defined architecture using Django.

3.1.2. Implement must-have features

3.1.3. Implement extra features

3.1.4. Implement PostgreSQL database following the defined structure.

3.2. Front-end

3.2.1. Develop user interface using React

3.2.2. Ensure user experience

4. Testing

4.1. Feature testing

4.1.1. Write test cases

4.1.2. Test defined features (functions)

4.2. User testing

4.2.1. Provide user experience feedback.

4.2.2. Test user accessibility

5. Presentation and Demo

5.1. Presentation

5.1.1. Prepare team's workflow

5.1.2. Prepare app's workflow

5.2. Demo

5.2.1. Demo core features

5.2.2. Demo extra features

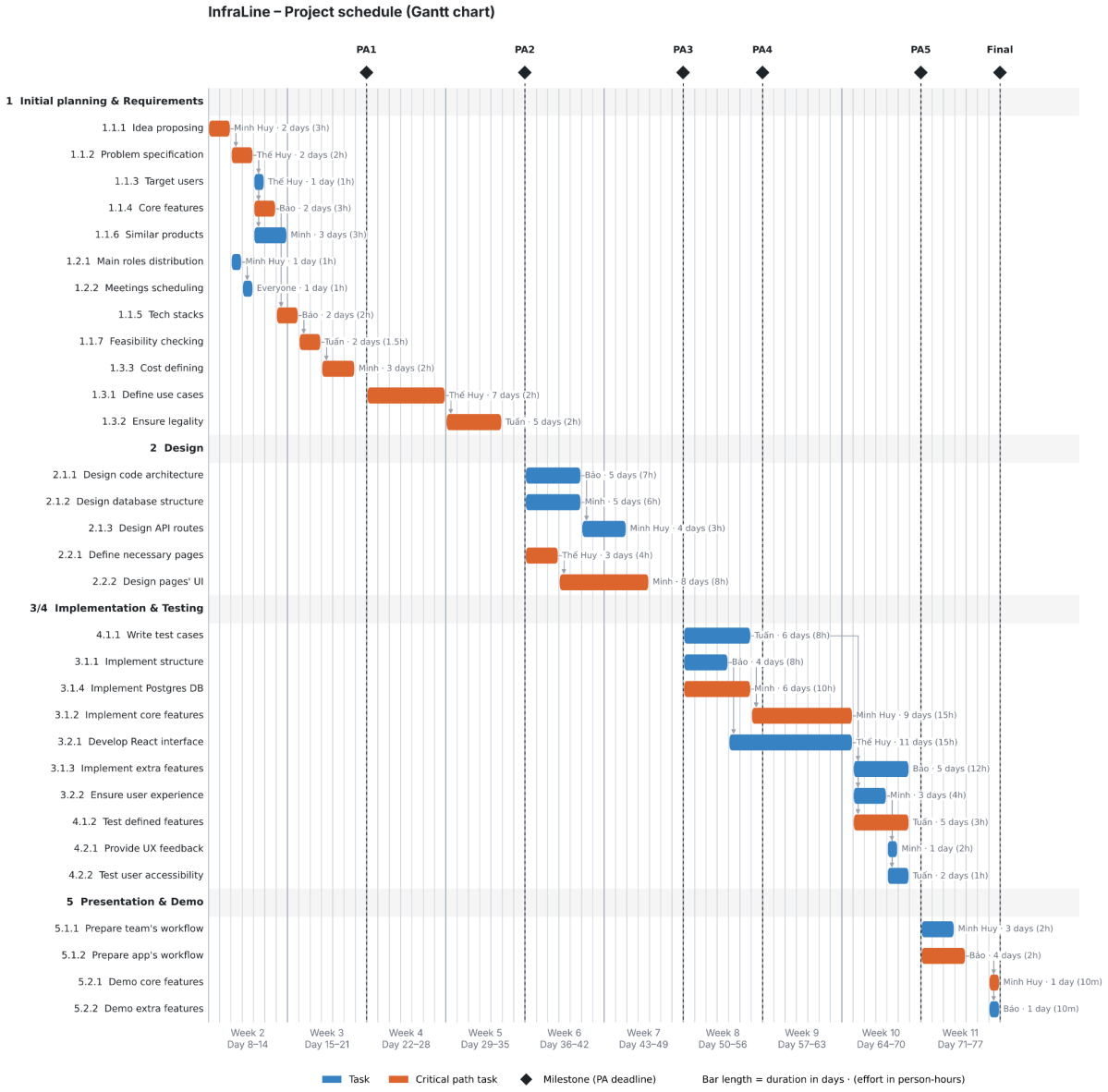
4.3 Estimation

WBS ID	Work package	Owner	Effort (person-hours)	Method / assumptions
1.1.1	Idea proposing	Hoàng Minh Huy	3 hours	Expert judgement: Inspired from real life problems.
1.1.2	Problem specification	Nguyễn Thế Huy	2 hours	Expert judgement: Researching current workflows
1.1.3	Target users	Nguyễn Thế Huy	1 hour	Expert judgement: Researching current workflows
1.1.4	Core features	Lê Nguyễn Quốc Bảo	3 hours	Three-point estimate: Researching possible features
1.1.5	Tech stacks	Lê Nguyễn Quốc Bảo	2 hours	Three-point estimate: Researching available technology based on features
1.1.6	Similar products	Nguyễn Chí Minh	3 hours	Expert judgement: About 1 hour for each of the 3 similar products.
1.1.7	Feasibility Checking	Đoàn Minh Tuấn	1.5 hour	Three-point estimate: discuss idea with developer
1.2.1	Main roles distribution	Hoàng Minh Huy	1 hour	Analogy: Experience from past projects
1.2.2	Meetings scheduling	Everyone	1 hour	Expert Judgement: Info of members' schedule acquired
1.3.1	Define use cases	Nguyễn Thế Huy	2 hours	Expert judgement: Researching industries and refining the scope of the project.
1.3.2	Ensure legality	Đoàn Minh Tuấn	2 hours	Three-point estimate: searching on web
1.3.3	Cost defining	Nguyễn Chí Minh	2 hours	Expert judgement: Checking free plans and pricing.
2.1.1	Design code architecture	Lê Nguyễn Quốc Bảo	7 hours	Analogy: Based on past projects in the Computational Thinking course.
2.1.2	Design database structure	Nguyễn Chí Minh	6 hours	Analogy: Based on past projects in the Database course.

2.1.3	Design API routes	Hoàng Minh Huy	3 hours	Three-point estimate: Find APIs + design routes
2.2.1	Define necessary pages	Nguyễn Thế Huy	4 hours	Three-point estimate: Reviewing specifications and discussing with developers.
2.2.2	Design pages' UI	Nguyễn Chí Minh	8 hours	Analogy: Based on past projects, about 1 hour per page for 8 pages.
3.1.1	Implement structure	Lê Nguyễn Quốc Bảo	8 hours	Analogy: Based on past projects in Computational Thinking and OOP courses.
3.1.2	Implement core features	Hoàng Minh Huy	15 hours	Three-point estimate: Implementation + Bug fixing + review
3.1.3	Implement extra features	Lê Nguyễn Quốc Bảo	12 hours	Analogy: Based on past projects in Computational Thinking and OOP courses.
3.1.4	Implement Postgres DB	Nguyễn Chí Minh	10 hours	Three-point estimate: Models + Migrations + Seed data
3.2.1	Develop React interface	Nguyễn Thế Huy	15 hours	Three-point estimate: Building and testing
3.2.2	Ensure user experience	Nguyễn Chí Minh	4 hours	Expert judgement: Reviewing the main user flows on desktop and mobile.
4.1.1	Write test cases	Đoàn Minh Tuấn	8 hours	Three-point estimate: many risks, huge tasks.
4.1.2	Test defined features	Đoàn Minh Tuấn	3 hours	Analogy: Based on past software projects
4.2.1	Provide UX feedbacks	Nguyễn Chí Minh	2 hour	Expert judgement: One feedback session and writing the feedback list.
4.2.2	Test user accessibility	Đoàn Minh Tuấn	1 hour	Three-point-estimate: discuss with BA.
5.1.1	Prepare team's workflow	Hoàng Minh Huy	2 hours	Analogy: Past experience
5.1.2	Prepare app's workflow	Lê Nguyễn Quốc Bảo	2 hours	Three-point estimate: Workflow + Setup + Fixing errors.
5.2.1	Demo core features	Hoàng Minh Huy	15 minutes	Three-point estimate: Setup + Internet connection.

5.2.2	Demo extra features	Lê Nguyễn Quốc Bảo	15 minutes	Three-point estimate: Setup + Internet connection.

4.4 Schedule



4.5 Roles and Responsibilities

Role	Main	Backup	Responsibilities
Team lead / Project manager	Hoàng Minh Huy	Lê Nguyễn Quốc Bảo	Plan, assign tasks, weekly status report, contact point
Business analyst	Nguyễn Thế Huy	Đoàn Minh Tuấn	Elicit and document requirements, organise reviews
Designer / Architect	Nguyễn Chí Minh	Nguyễn Thế Huy	Architecture, class, data and UI design

Developers	Lê Nguyễn Quốc Bảo	Hoàng Minh Huy	Implement, unit test, review pull requests
Tester	Đoàn Minh Tuấn	Nguyễn Chí Minh	Test plan, test cases, system testing, test report

4.6 Human Resources and Cost Estimate

✔ Total effort: 134 person-hours

✔ Hourly rate: 91.000đ (indeed, links:

https://jobs.vn.indeed.com/career/junior-developer/salaries/Thành-Phố-Hồ-Chí-Minh)

✔ Host: 0đ (Vercel, Render + Gunicorn)

✔ Database: 0đ (Supabase)

✔ Total cost (Estimated): 134 * 91.000 + 0 = 12.194.000đ

5 Risk Management

ID	Risk	Type	Probability	Loss	RE	Mitigation / contingency	Owner
R1	Spam and duplicate incident reports	Requirements	4	3	12	Display a list of "Processing" incidents in that room directly on the screen before sending.	Lê Nguyễn Quốc Bảo
R2	Schedule conflict with midterm exams	Organisational	4	4	16	Divide work to all members effectively.	Hoàng Minh Huy
R3	Database slows down when many users report at once	Technology	2	4	12	Limit requests per user and optimize database queries.	Lê Nguyễn Quốc Bảo
R4	Not enough time left to test features before the deadline	Estimation	4	5	20	Prepare a test plan based on past projects to apply to the main project for time saving.	Đoàn Minh Tuấn
R5	Loss of data due to lack of backups.	Technology	2	4	8	Write a script to automatically load seed data for immediate restoration when needed; configure automatic backups for the database.	Nguyễn Chí Minh
R6	Key member is sick or unavailable during PA5	People	2	5	10	Share knowledge across the team through communication channels and keep all code updated on Git.	Hoàng Minh Huy
R7	Ambiguous ticket completion criteria	Requirements	4	2	8	Technicians are required to attach photos of the completion inspection after repairs. Add an additional step for the reporter to test it for around 3-4 days to confirm satisfaction before the ticket is fully closed.	Nguyễn Thế Huy
R8	Refusal of cooperation from team member	People	2	5	10	Strictly follow the pre-defined team contract to avoid conflict or resolve it peacefully.	Hoàng Minh Huy

6 Tools & Collaboration Setup

One of the main goals of the project is to practise teamwork for professional software development. The use of tools is monitored and graded (teamwork component). The team must use:

- ✔ Moodle – announcements and PA submissions.
- ✔ Communication – Discord
- ✔ Task board – GitHub Projects
- ✔ Git repository – GitHub:
  - main branch is protected; changes are merged through pull requests with at least one review
  - README (description, setup), .gitignore; never commit secrets (.env, API keys, passwords)
  - commit messages follow the Conventional Commits style (feat:, fix:, docs:, test:, ...)
- ✔ Weekly status meeting (15 minutes) and weekly status report (template provided), committed to /docs/management/weekly every week from week 2 to week 10.
- ✔ Weekly deadline: Tuesday 11 P.M.

Repository structure:

Folder	Content
/src	Source code
/docs/management	Project plan; weekly status reports in /docs/management/weekly
/docs/requirements	Vision, SRS, use cases, mockups
/docs/design	Architecture (C4), ADRs, UML models, UI design
/docs/test	Test plan, test cases, test reports
/docs/ai-usage	AI usage logs
/pa/PA1 ... /pa/PA5	One sub-folder per PA submission (PDF + source files)

7 AI Usage Log

#	Member	Tool	Task / purpose	How the output was verified and changed
1	Lê Nguyễn Quốc Bảo	Claude Sonnet 5.5	Researching the options for hosting / deployment as well as limits on the free plan for those options	Verified by picking the option that fits our project the most and manually checking their pricing / limits on the free plan.
2	Lê Nguyễn Quốc Bảo	Claude Sonnet 5.5	Researching the available services and API for email response	Create an account on Resend and Mailjet and check out their limits and pricing.
3	Nguyễn Chí Minh	Gemini	Researching what a C4 System Context diagram is and which elements it needs	Verified by checking the official C4 model website and drawing our own diagram for InfraLine in draw.io.
4	Nguyễn Chí Minh	Gemini	Asking whether Supabase should be an external system in the context diagram	Verified with the C4 model definition and leaving Supabase out of the Level 1 diagram.
5	Nguyễn Chí Minh	Claude Opus 5.5	Drawing the Gantt chart and identifying the critical path in Section 4.4	Checked that every task, owner and effort matches the WBS and Section 4.3, and that the milestones match the PA deadlines in Section 4.1. The schedule and critical path were also reviewed by Hoàng Minh Huy (team lead).
6	Đoàn Minh Tuấn	Gemini Argon	Give some example of 5 risks of InfraLine project	Give me each risk 5 examples for me to check and put to the report.